

RUTIKA AVINASH KADAM

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SUMMARY

Data science and AI professional with MSc, Data Science and 4+ years of experience building scalable ML and generative AI systems with a focus on rigorous evaluation and measurable impact. Collaborative problem-solver and clear communicator.

EDUCATION

Stony Brook University

Master of Science in Data Science

Stony Brook, USA

August 2024 – May 2026

Coursework: Machine Learning, Deep Learning, Natural Language Processing, Large Language Models, Agentic AI, Big Data.

Savitribai Phule Pune University

Bachelor of Engineering in Information Technology

Pune, India

August 2016 – May 2020

Coursework: Statistics, Discrete Structures, Data Structures & Algorithms, System Design, Cloud Computing.

SKILLS

Programming Languages: Python, R, SAS, SQL (MySQL, PostgreSQL, PL/SQL), C++, HTML, CSS

Data Science: NumPy, Pandas, Seaborn, scikit-learn, TensorFlow, Keras, PyTorch, Hugging Face, LangChain, LangGraph

Data Engineering & Big Data: Azure Data Factory, Data Lake, Databricks, Microsoft Fabric, SSMS, SSIS

Cloud & DevOps: Azure, FastAPI, REST APIs, Docker, Kubernetes, MLflow, Git, GitHub, Postman

Visualization & Analytics: Power BI, Tableau, Microsoft Excel

Security & IT Operations: Qualys VMDR, Microsoft Endpoint Configuration Manager (SCCM), ServiceNow, Jira, Aternity

PROFESSIONAL EXPERIENCE

Stony Brook Medicine

Data Scientist (Graduate Researcher)

Stony Brook, USA

July 2025 – Present

- Built an end-to-end **multi-modal RAG pipeline** over **6k+ pages of medical documents** for the Department of Public Health, using **Azure Document Intelligence** for layout-aware extraction and **GPT-4o** for figure captioning.
- Implemented **custom semantic chunking** using embedding-based sentence similarity and adaptive breakpoint detection.
- Designed & benchmarked **FLAT, HNSW, & IVF_FLAT indexes in Milvus** across **27k+ multi-modal chunks**, comparing p50/p95 latency & Recall@5; achieved comparable accuracy with approximate indexes at **~2.5× lower latency** than FLAT.
- Evaluated **vector search, MMR reranking, and hybrid Vector+BM25 retrieval with RRF** on a synthetic ground-truth dataset spanning text, tables, and images; achieved **0.90 Recall@5** with hybrid retrieval.
- Built a **grounded RAG generation pipeline** with source citations, measured **RAGAS scores of 0.82 faithfulness, 0.82 answer relevancy, 0.93 context precision, and 0.74 context recall**.

Tata Consultancy Services

System Analyst

Pune, India

August 2020 – April 2024

- Performed **exploratory data analysis** on **1M+ system vulnerability records** reporting from **Qualys VMDR** using **Python-pandas**, uncovering trends, anomalies, and threat vectors, that informed remediation prioritization.
- Built **linear and logistic regression models** (scikit-learn) predicting CVSS severity and exploit availability on **150K findings**, achieving **0.74 ROC-AUC** and a **22% risk-exposure reduction** via backtest vs. FIFO prioritization.
- Discovered recurring remediation patterns across **150K+ detection records** by applying **semantic clustering using E5 embeddings with HDBSCAN** on remediation text data.
- Fine-tuned a **DistilBERT classifier** on cluster-derived labels to predict remediation categories, achieving **100% held-out accuracy**, laying the groundwork for automated ticket routing.
- Achieved 99% patch compliance across enterprise systems by deploying automated remediation workflows through Microsoft Endpoint Configuration Manager and automated tasks using PowerShell, boosting productivity by 25%.

Zensar Technologies

Data Intern

Pune, India

May 2019 – July 2019

- Implemented **ETL pipelines in Azure Data Factory with incremental batch loading** to ingest **multi-channel website traffic and operational data** into **Azure Data Lake Gen2** for scalable data integration for **Swiggy-India**.
- Developed **Power BI dashboards** to visualize **conversion funnel metrics and marketing channel attribution**, identifying drivers of traffic variability and campaign performance, hence increasing **high-intent traffic by 15%**.

PROJECT EXPERIENCE

Electric Vehicle Battery Capacity Prediction | PyTorch, MLOps, Data Version Control, Azure

- Estimated **EV battery capacity** from ordinary charging data, since direct measurement needs controlled conditions present in only **56%** of real sessions; trained **PyTorch** models on **111K** charging windows from 30 vehicles with vehicle-wise **5-fold CV**.
- Benchmarked **LSTM** and **Transformer** against simple baselines across **10 tracked experiments**; a linear model on odometer alone won (**RMSE 1.10** vs 1.69/1.71), and a **positional-encoding ablation** showed the 21-minute window, not model capacity, was the limit.
- Versioned data and models in **DVC** across two cloud remotes, tracked experiments in **MLflow**, and shipped a **FastAPI** service via **Docker** and **Azure Container Registry** to **Azure Container Apps** on a public HTTPS endpoint, with **CI** running **43 tests** per push.